Lab 03: Bash Scripting

Lab 3: Bash Scripting

Author

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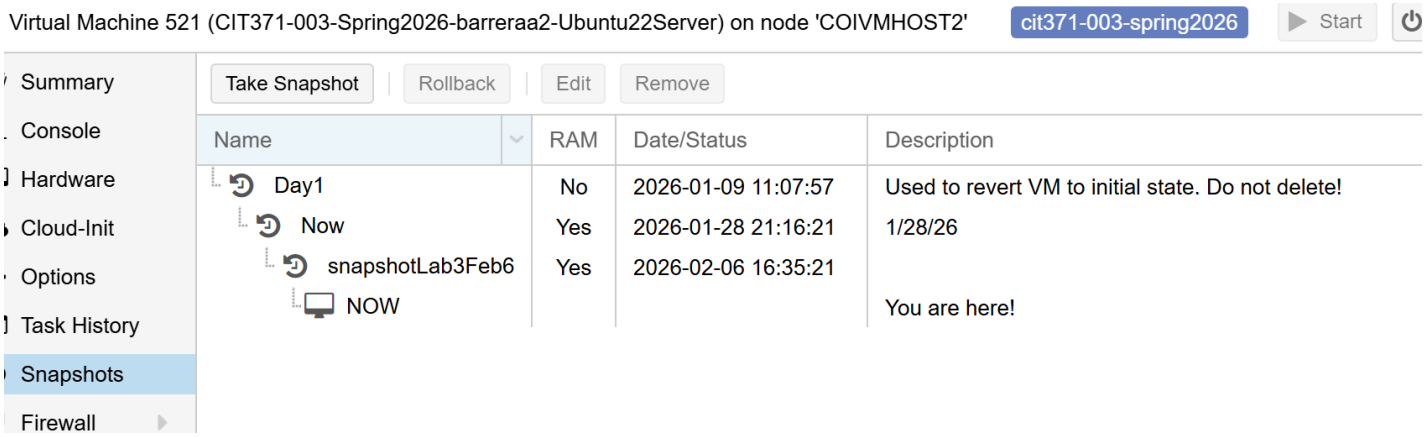
VM Snapshot(s)

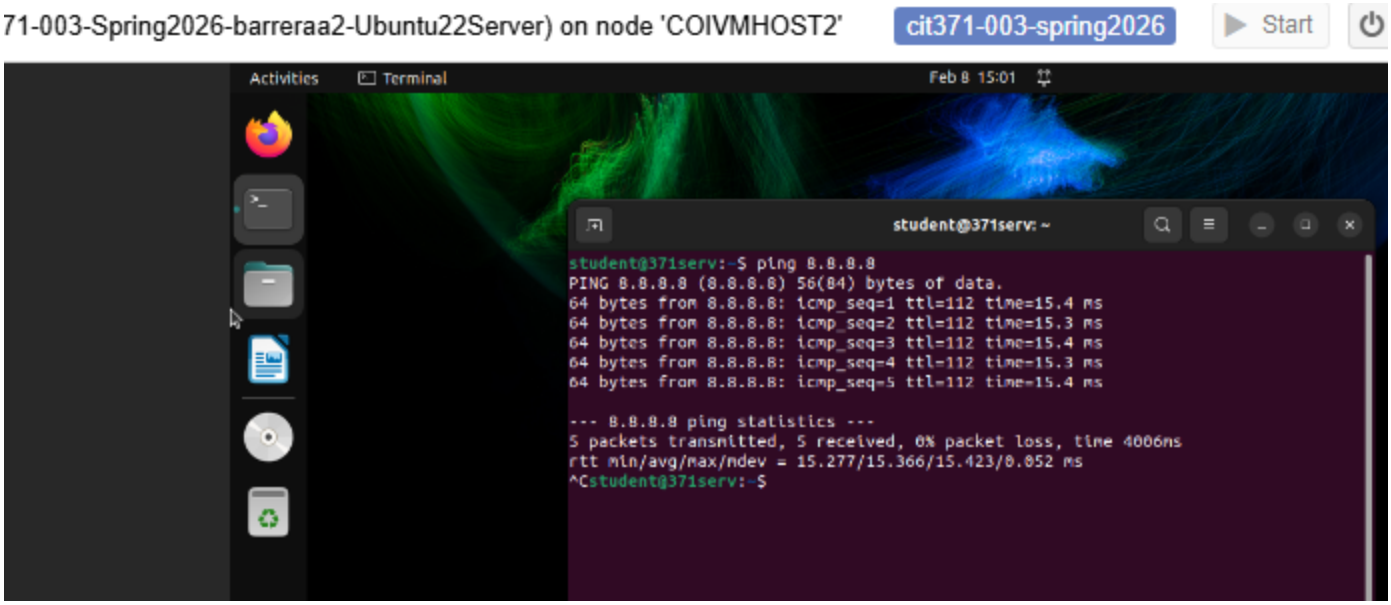


Overview

(Insert overview text and screenshots)

Part 1

We will need to ping 8.8.8.8 to test network connectivity.



Part 2

In this part we will need to write a Bash script that will help us create a users using the “add user command.” The script will need to ask us first information variables about the user before we run the script.

It should are the following:

- Login Name ,
- GECOS / Full Name ,
- Login Shell,
- Home directory .

This script should also have status of the comments and messages.

The format we will need to create this script is by first printing a message that says “Retrieving info to create new user" Then it should ask for each of the account details above providing information on what details will go to each variable.

Then after all variables are provided the bash script should print a message stating “Creating User…” then it should provide a message when it finally created it showing the login name with the login variable.

First we will create the file and we will call it “creating_user.sh” using the nano command as it shows below in the terminal.

```
nano creating_user.sh
```

This will bring up to the nano area where we will write our script.

To make the script executable we will need to type this command inside known as shebang, it let's the system know to use bash interpreter to run the script.

```
#!/bin/bash
```

Then we will start asking the user for information below:

- In the below command we are initializing the status message for the script.

```
echo "Retrieving info to create new user."
```

- Then we are going to ask the user for their login name and add that information to the variable LOGIN_NAME

```
echo "Please enter a login name"
read LOGIN_NAME
```

- In the command below we are asking for the full name and add that information to the variable FULL_NAME

```
echo "Please enter the user's full name (GECOS)"
read FULL_NAME
```

- This next command we are going to ask for the login shell (for example: /bin/bash) and put that in the variable LOGIN_SHELL

```
echo "Please enter the login shell (example: /bin/bash)"
read LOGIN_SHELL
```

- Lastly we are going to add for the home directory and add that to variable HOME_DIR

```
echo "Please enter the home directory (example: /home/username):"
read HOME_DIR
```

- In the next steps we will provide information that we are creating the user with the details that were given…

```
echo "Creating user..."
```

- In the command below we are going to start creating the users with the information that is in the variables..

```
adduser \
--gecos "$FULL_NAME" \
--shell "$LOGIN_SHELL" \
--home "$HOME_DIR" \
"$LOGIN_NAME"
```

- In this final message we will let the person know that their user has been created with the information they provided and showing them the Login name of the account.

```
echo "User account 'LOGIN_NAME' has been successfully created."
```

Now that we have the script we will need to test it. We will be to sign in using root with the “sudo -i” command. This root permissions are needed to run the “adduser” command.

To run the script we will need to go to the directory where the script is saved by putting this command:

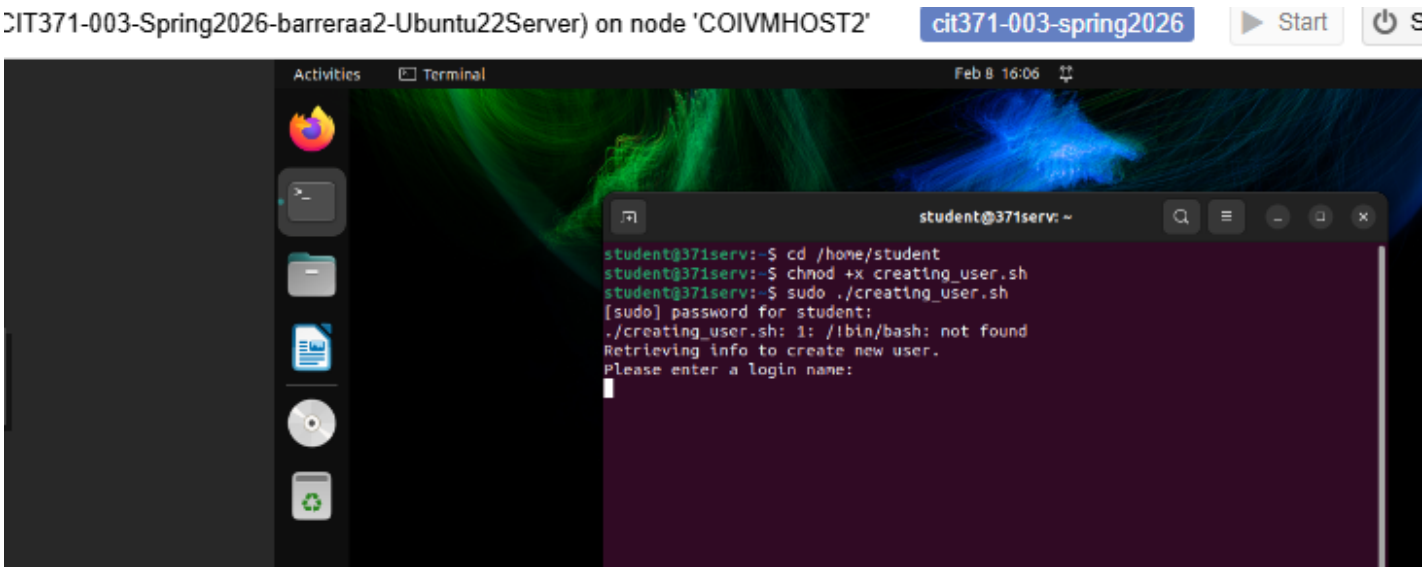
```
cd /home/student
```

Then we will make sure is executable with this command:

```
chmod +x creting_user.sh
```

And then we will run it with this command:

```
sudo ./creating_user.sh
```



No we see that the script is running and asking for information.

We will provide a new user to be added with this information below:

Login name: mcben1

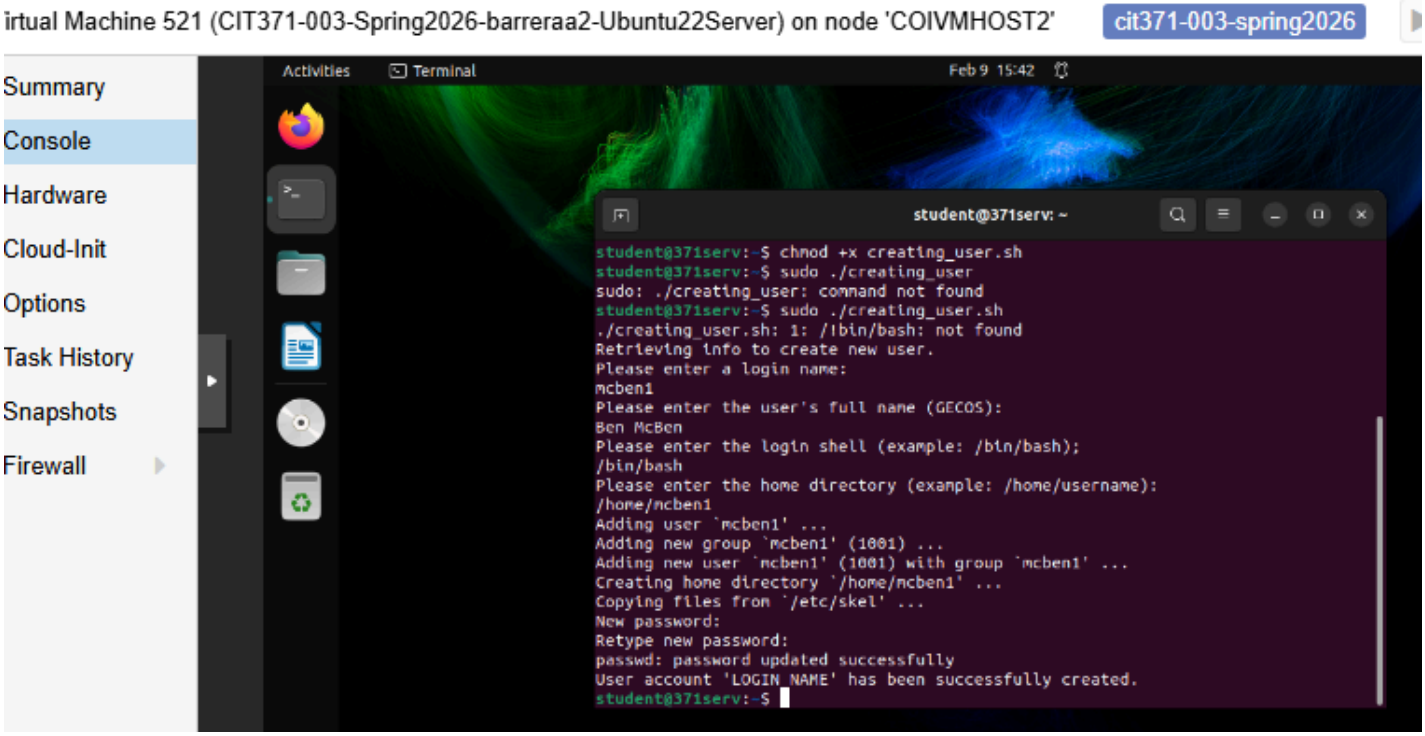
GECOS /Full name: Ben McBen

Login shell: /bin/bash

Home directory: /home/mcben1

Password: Student371!

You will provide the password whenever the adduser account requests one and it will not be giving to a variable.



Part 3: Directory Contents Listing Script

We will now write a second bash script , in this script we will check the contents of the current working directory and print what the names of the files and directories that are currently in it. This will print all the file names for one group and the directory names in another. We will now get out from the mcben1 account clicking exit twice and then opening the terminal again as the student user.

We will write the script using the nano command and then saving the script with the name we want. Here is the command below to start writing the script.

```
cd /home/student
nano contents.sh
```

Then it will bring up the nano to write the script. We want to store the contents of the current directory in a variable so we will go ahead and add this command and also the essential command when writing a bash script.

```
#!/bin/bash
contents=$(ls)
```

We will now check for files in this first iteration.

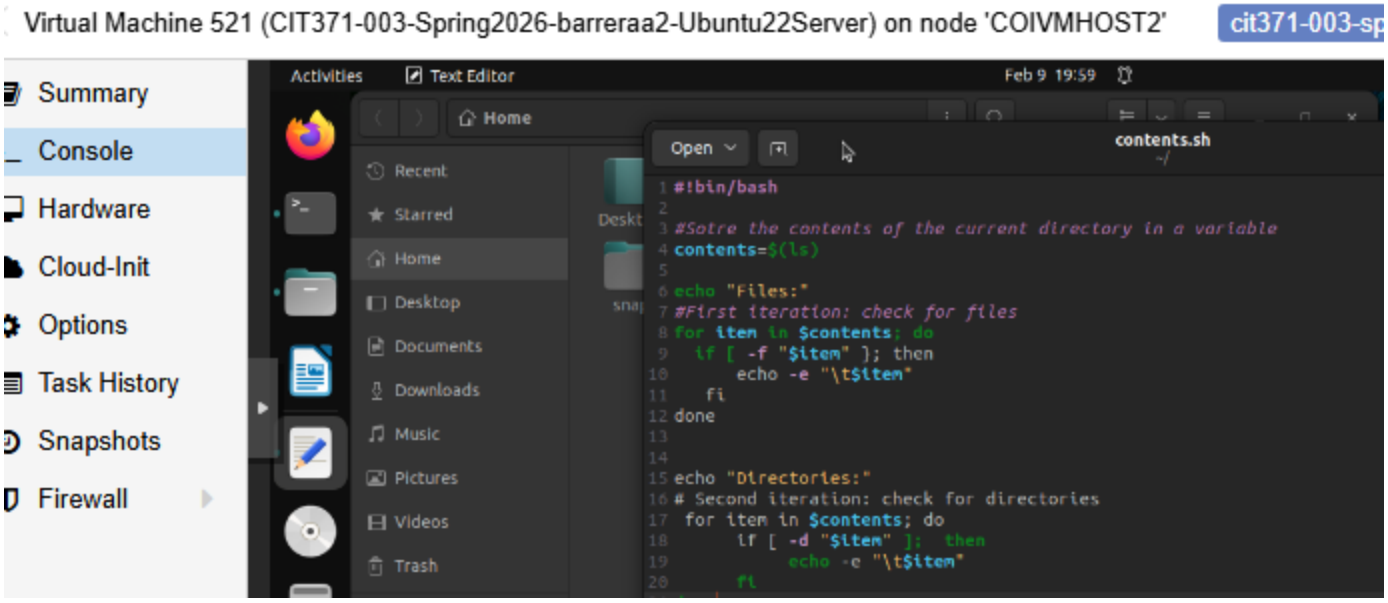
Put this code to check the files:

```
echo "Files:"
for item in $contents; do
   if [-f "$item"]; then
       echo -e "\t$item"
   fi
done
```

Now in this second part we will check for directories, using this code:

```
echo "Directories:"
for item in $contents; do
     if [ -d "$item" ]; then
          echo -e "\t$item"
     fi
  done
```

After we have this code, we will need to save it with Ctrl + X and then we click on the Y and then press enter and it will save with the name we give it which it "contents.sh"



Now that we have the script we could run it in the home/student where we created it. In the directory we run it it should have multiple files and directories so the script could show us all the information.

First we need to move to the directory where we have the bash script. using this command:

```
cd /home/student
```

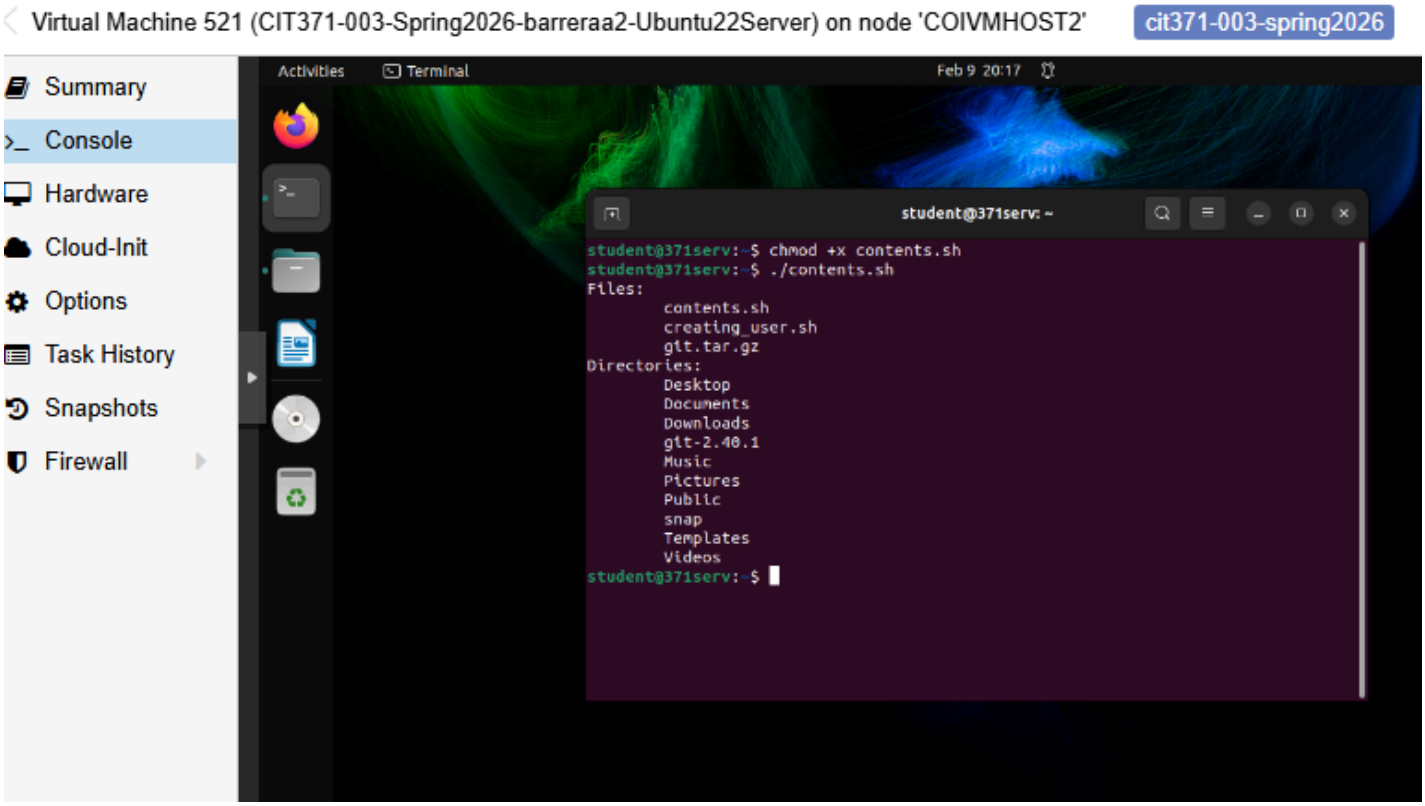
Then we will make our Script executable by putting this command in our command line:

```
chmod +x contents.sh
```

Then we will run our script:

```
./contents.sh
```

Below is the screenshots we see of the files and directories inside that home/student.



Reference

Google: Google.com

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